



## A healthier water use strategy in primitive forests contributes to stronger water conservation capabilities compared with secondary forests

Zeyu Sun<sup>a,c,d,1</sup>, Mingming Feng<sup>a,b,c,1</sup>, Xinyan Zhang<sup>d</sup>, Shaoqing Zhang<sup>a,c</sup>, Wenguang Zhang<sup>a,c,\*</sup>, Yang Li<sup>a,c,e</sup>, Yiqiang Huang<sup>a,c,e</sup>, Peng Qi<sup>a,c</sup>, Wenjuan Wang<sup>a,c</sup>, Yuanchun Zou<sup>a,c</sup>, Ming Jiang<sup>a,c,\*</sup>

<sup>a</sup> Key Laboratory of Wetland Ecology and Environment, Northeast Institute of Geography and Agroecology (IGA), Chinese Academy of Sciences (CAS), Changchun 130102, PR China

<sup>b</sup> University of Chinese Academy of Sciences, Beijing 100049, PR China

<sup>c</sup> Jilin Provincial Joint Key Laboratory of Changbai Mountains Wetland and Ecology, Changchun 130102, PR China

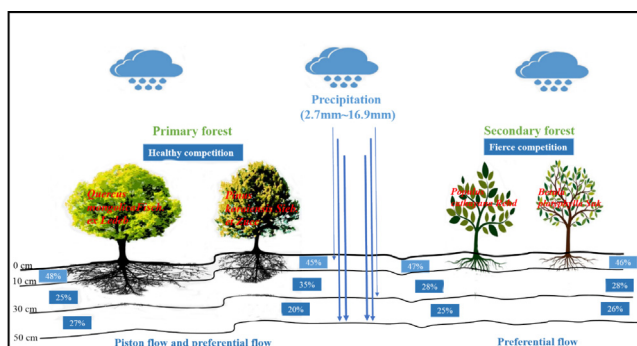
<sup>d</sup> School of Chemistry and Environmental Engineering, Changchun University of Science and Technology, Changchun 130022, PR China

<sup>e</sup> College of Geography and Ocean Science, Yanbian University, Yanbian 133002, PR China

### HIGHLIGHTS

- The primary forest with a stronger water conservation capacity than that secondary forest.
- Primary forest with a healthy competition, but secondary forest with a fierce competition.
- The soil water in primary forest was in line with the theory of a soil reservoir, while the secondary forest was according with the hypothesis of "two water worlds"

### GRAPHICAL ABSTRACT



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### ABSTRACT

Water conservation is an important ecological function of forest ecosystems, plant water use strategy is a key factor in regulating forest ecosystem water balance. However, there are still insufficient studies on the water conservation capacity and water use strategies of different forest types, especially in climate-sensitive areas. In this study, we determined the stable isotope values ( $\delta D$ ,  $\delta^{18}O$  and  $d$ -excess) of plant water, soil water and precipitation from two typical stand types (primary forest and secondary forest) on Changbai Mountain to reveal plant water use and evaluated the water conservation capacity. The results indicated that rainwater infiltrated into the soil combined with piston flow and preferential flow in the primary forest, and preferential flow was the only form of flow in the secondary forest. The main tree species in the primary forest formed a relatively stable water use niche. Among them, the water use pattern of *Quercus mongolica* Fisch. ex Ledeb (Qm.) was transformed between shallow and deep soil layers with strong ecological plasticity. The dominant species in secondary forest derived water from similar soil layers with intense interspecific competition. By comparing the water use patterns, the secondary forest conformed to the hypothesis of "two water worlds", while the primary forest conformed to the hypothesis of one reservoir. The primary forest ecosystem had stronger water conservation capacity than secondary forest ecosystem due to the regulable water use strategies of plants and the stable water conservation capacity of the soil. These results will provide theoretical support and a reference for plan future forest management strategies in the climate-sensitive areas.

\* Corresponding authors at: Wenguang Zhang, Key Laboratory of Wetland Ecology and Environment, Northeast Institute of Geography and Agroecology (IGA), Chinese Academy of Sciences (CAS), Changchun 130102, PR China.

E-mail address: [zhangwenguang@iga.ac.cn](mailto:zhangwenguang@iga.ac.cn) (W. Zhang).

<sup>1</sup> The author contribute equally in this study.

## 1. Introduction

Water conservation can be expressed as the processes and ability of ecosystems to maintain water within a certain time and space (Lu et al., 2015). As an important ecological service function, water conservation is affected by the interaction of the soil-vegetation-atmosphere system (Manoli et al., 2016; Sprenger et al., 2017; Zhang et al., 2017; Zhang et al., 2019a). Plants are the link for water circulation between the soil and atmosphere. On the one hand, they absorb water through transpiration to adjust the water balance between plants and the surrounding environment (Deng et al., 2020; Gao et al., 2017). On the other hand, plant roots can redistribute water to change the soil water content and affect the growth of the plant community according to climate change impacts such as drought, intense heat and flood (Burgess et al., 1998; Zhang et al., 2021). Therefore, it is important to explore plant water use strategies to better understand the water conservation function of forest ecosystems.

Climate change has accelerated the global water cycle, aggravating extreme climate events (eg. heavy rainfall and heat waves) and resulting in an imbalance in water distributions in different areas (Stocker et al., 2014). Under extreme climate events, the intra and interspecific competition of forest ecosystems could be aggravated and result in high mortality of tree species (Allen et al., 2010; Kelly and Goulden, 2008; Luo and Chen, 2013; Zhu et al., 2014). However, forests have different capacities to mitigate climate change due to water interception, storage and transfer to varying degrees (Hou et al., 2011; Liu et al., 2021b). Therefore, it is of great significance to explore the relationship between forest types and water conservation capacity and to construct scientific management strategies.

Plant water use refers to the different water absorption modes of plants to adapt to the environment under changes in time and space (Barbier et al., 2006; Dai et al., 2020; Robertson and Gazis, 2006). Studies have shown that in arid and semiarid areas, plants use deep water sources (e.g. deep soil water and groundwater) during the dry season and obtain shallow water sources in the rainy season (Asbjornsen et al., 2008; Dawson and Pate, 1996). Drought is the main cause to the water demand of plants exceeded the supply of soil water, which could affect the plant community structure and ecosystem processes including trees with weak adaptability growth slow down or even die, the forest ecology services function was fully weakened (Chen et al., 2015). In contrast, precipitation increases soil moisture and promotes plant growth (Aspinwall et al., 2017; Keenan et al., 2013), but high water content could inhibit the water absorption of plant roots and may also lead to the cessation of plant growth or even death (Henri, 2001; Sajedi et al., 2012). Therefore, to promote species coexistence and ecosystem stability, healthy water use strategies should be formed among plant species (Eggemeyer et al., 2009; Wu et al., 2016; Silvertown et al., 2015).

Studies have shown that most plants, with the exception of some dry and saline plants (Barbeta et al., 2019; Ellsworth and Williams, 2007), do not undergo root uptake (Dawson and Ehleringer, 1993) and transpiration (Martin-Gomez et al., 2017) during isotope fractionation (Brunel et al., 1995; Dawson et al., 2002). Therefore, the stable isotopic composition of plant xylem water can be considered the isotopic composition of its water source (Zhao et al., 2016), and stable isotopes can be an effective method to discern the source of plant water use when there are differences in the stable isotopic composition of each potential water source (Ehleringer et al., 1991; Swaffer et al., 2014; Yang and Fu, 2017).

The primitive broad-leaved Korean pine forest (called it as primary forest at following study) of the Changbai Mountain is the largest temperate forest ecosystem in China (Zheng et al., 2019a). In the middle of 19th century, it had suffering from broadscale cutting down. The vegetation in the deforested area has undergone half a century of development and succession and now forms a secondary poplar-birch forest (called it as secondary forest at following study). These two stand types together compose the temperate forest ecosystem of the Changbai Mountain. The stand age and canopy density of poplar-birch forest are less than broad-leaved Korean pine forest. Therefore, we hypothesize that primary forest should have a

stronger water conversation capacity and a better water use strategy than secondary forests. Therefore, we collected soils and xylem water samples of dominant species in two forest types during precipitation events and measured their hydrogen and oxygen stable isotopes value to analyze the water conversation and water use strategy of those stand types. The research objects are to: 1) quantify the plant water use pattern in the two forest types during the different rainfall events; 2) evaluate the water conservation capacity of the two forest types.

## 2. Materials and methods

### 2.1. Study area

The study area is located in the natural forest area on the northern slope of Changbai Mountain (Fig. 1, Table 1), which is at the northeastern edge of the global monsoon climate, with a temperate humid monsoon climate influenced by continents (Hou et al., 2012). The average annual precipitation and temperature from 2005 to 2020 were 691 mm and 4 °C, respectively, with 77 % of precipitation concentrated during the growing season (May to September) (Fig. 2A). Changbai Mountain acts as the source of the Songhua River, Tumen River, and Yalu River. It is an important water tower, hydropower energy base, and source of high-quality natural mineral water in Northeast China (Qi et al., 2021). The forest in Changbai Mountain is one of the three major primary forests in China and serves as a key reservoir of water for this region.

The primary broadleaf Korean pine forest is the regional vegetation in the Changbai Mountain area; with a mean canopy height of 26 m. A dense understory, consisting of multispecies broad-leaved shrubs, has a height of 0.5–2 m. The peak leaf area index is approximately 6 (Zhang et al., 2006). Disturbed by human activities and natural disasters, some of the primary forests have formed secondary forest communities through secondary succession, among which the typical secondary forest is dominated by *Populus davidiana* (Po.) and *Betula platyphylla* Suk (Bp.) (Tian et al., 2021). The soil type in the study area is albic dark brown forest soil that developed on the basalt platform (Zhang, 1987; Sun et al., 2016; Dong et al., 2019), which is formed through white-pulpification and other soil-forming processes with a dark humus surface layer and an off-white subsurface layer. The soil layers of the two stands are the humus eluvial horizon (approximately 10 cm), albic soil layer (approximately 15 cm) and sediment layer (approximately 45 cm) (Zhao et al., 2019).

### 2.2. Sample collection

Before and after each precipitation events from May to September 2020, a total of 571 water samples were collected, including 82 precipitation samples (all rainfall events), 216 plant xylem samples and 273 soil water samples (three rainfall events, light rainfall, mid-rainfall and continuous rainfall, three replicates of each tree species and plots). Precipitation samples were collected using plastic funnel bottle sets immediately after each rainfall event, and then stored at approximately 2 °C to 4 °C. On the basis of the vegetation plot survey results, *Quercus mongolica* Fisch. ex Ledeb (Qm.), *Pinus koraiensis* Sieb. et Zucc. (Pk.), *Tilia amurensis* Rupr (Ta.), *Fraxinus mandshurica* (Fm.) and *Acer pictum* Thunb. ex Murray (Am.) were selected, and Qm., Pk. Ta., *Populus davidiana* (Po.), *Larix gmelinii* (Rupr.) Kuzen. (Lg.), *Betula platyphylla* Suk (Bp.) were selected for the secondary forest. For each species, approximately 4 xylem strips with a length of 4–5 cm were collected into 8 ml brown glass bottles.

The characteristics of soil development at each vegetation plot were recorded. Soil samples were collected from each depth section of 0–5 cm, 5–10 cm, 10–20 cm, 20–30 cm, 30–40 cm and 40–50 cm and placed in 8 ml brown 147 glass bottles. All vegetation and soil samples were sealed with Parafilm immediately after collection and transported back to the laboratory quickly in a refrigerated box with ice packs, with plant and soil samples being kept frozen at –25 °C and water samples kept refrigerated.

In addition, the soil samples were also collected in Ziplock™ bags, air dried, then ground in order to pass through a 100-mesh sieve. The soil

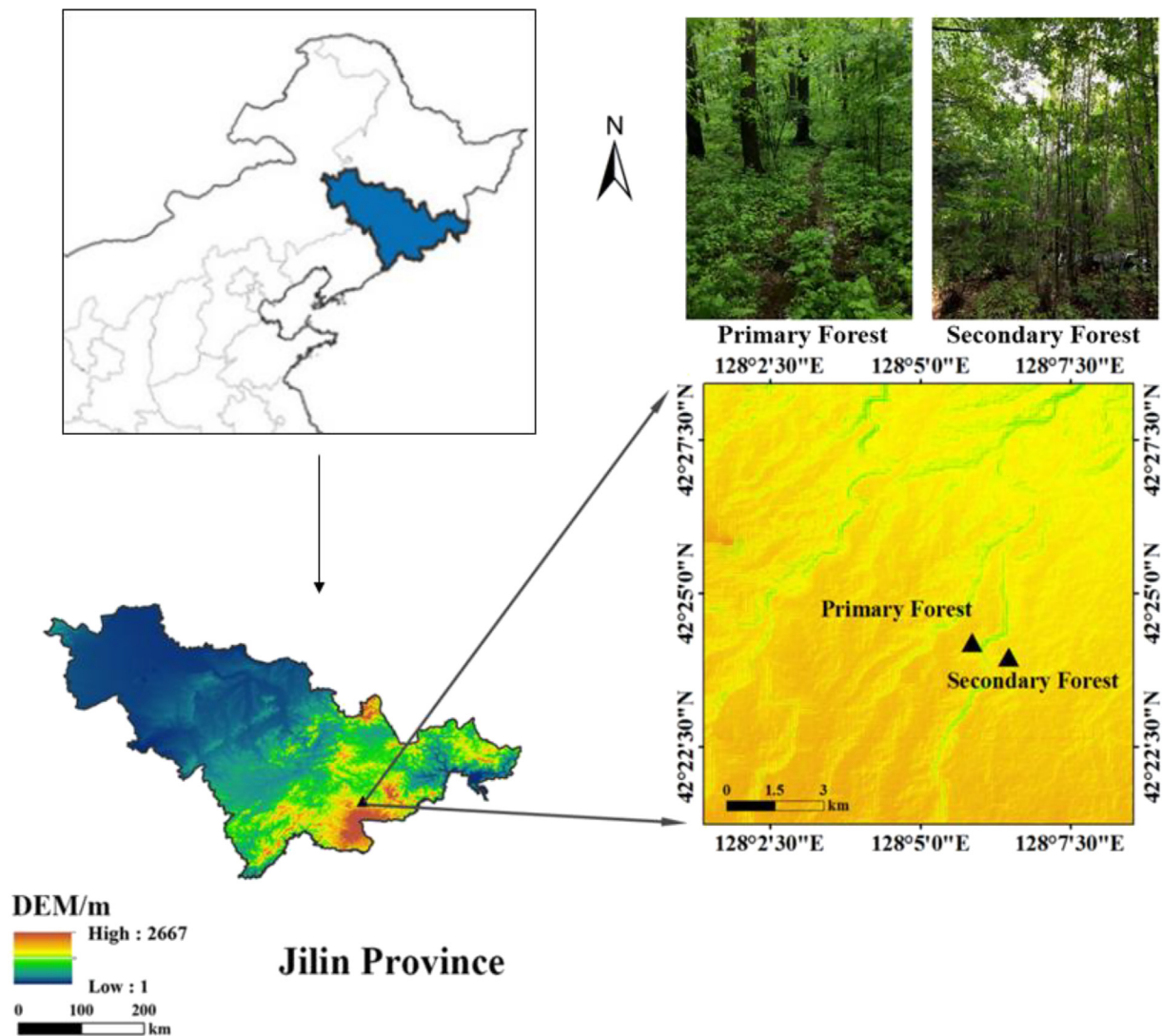


Fig. 1. Location of study area, including primary forest and secondary forest in Jilin Province, China.

bulk weight was calculated using a cutting ring, and other physicochemical properties were determined at the Center of Forecasting and Analysis of Northeast Institute of Geography and Agroecology, Chinese Academy of Sciences (CFA, IGA, CAS).

The water of the soil and vegetation samples was first extracted by a low-temperature vacuum extraction device with a volume within 0.2–2 ml and then loaded into an automatic sample bottle. Last, all the water samples collected were analyzed for  $^{18}\text{O}/^{16}\text{O}$  and  $^2\text{H}/^1\text{H}$  ratio using a Thermo Fisher MAT253 isotope mass spectrometer coupled with Elemental Analyzer Flash 2000 HT (Pyrolysis through high temperature EA) in the CFA, IGA, CAS. The results were expressed in the  $\delta$  notation as differences in parts per thousand relative to the Vienna Standard Mean Ocean Water (VSMOW). The measurement precisions for  $\delta\text{D}$  and  $\delta^{18}\text{O}$  were 0.5 ‰ and 0.2 ‰, respectively.

**Table 1**  
Stand characters of the primary forest and secondary forest.

| Forest type      | Longitude and latitude      | Altitude (m) | Standage | Closure | Treeheight (m) |
|------------------|-----------------------------|--------------|----------|---------|----------------|
| Primary forest   | N42°24'12"<br>E 128°05'52"  | 759          | >250     | 0.85    | 26–28          |
| Secondary forest | N 42°23'58"<br>E 128°06'28" | 738.1        | 30–50    | 0.7     | 18–20          |

### 2.3. Data analysis

The MixSIAR model was used to determine soil moisture and plant water sources (Brunel et al., 1995; Zhang et al., 2020). Compared with the traditional end-member mixing model, MixSIAR contains uncertainty analysis in the estimate process and print results with higher accuracy. The formulation of this model is as follows:

$$\phi_i = i|\gamma(P_i) = v^T \log \left[ \frac{p_{i1}}{g(P_i)}, \dots, \frac{p_{iK}}{g(P_i)} \right] \text{ with } g(P_i) = \left( \prod_{k=1}^K p_{ik} \right)^{1/K} \quad (1)$$

In this formula,  $P$  is the proportion of each source. The inverse transformation  $P_i = i|\gamma^{-1}(\phi_i)$  simply involved exponentiating and renormalizing the values.  $p_{ik}$  is the contribution of source  $k$  for consumer  $i$ .  $p_i$  is the  $K$ -vector of source proportions for consumer  $i$  (Parnell et al., 2013).

The model factors including potential variability in isotopic values, isotopic fractionation, and noncoherence of multiple sources, as well as additional modules for input forms of raw multiple isotope data from contributing sources, random effects categorical variables, and residual-process errors, which enhanced the accuracy of calculations (Du et al., 2018; Zhang et al., 2020). First, the raw xylem isotope ratios ( $\delta\text{D}$  and  $\delta^{18}\text{O}$ ) were used as mixture data and uploaded to the MixSIAR model. Second, both the mean values and standard errors of isotope ratios ( $\delta^{18}\text{O}$ ) from each soil layer were treated as initial data and added into the model.



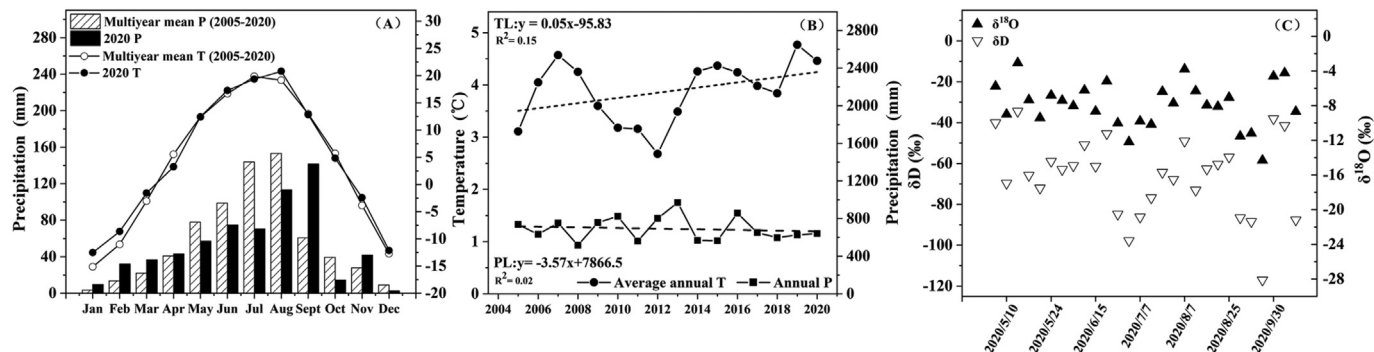


Fig. 2. The temporal variation in mean precipitation (P), temperature pattern (T) and isotopic composition of rainwater. (A) The monthly variation of multiyear mean precipitation (2005–2020), temperature and precipitation in 2020; (B) annual mean temperature and precipitation variation in Chingbai Mountain. (C) The temporal variation of isotopic compositions in rainwater. The linear (solid line) and fitting trend (dashed line): TL represents the temperature change and PL refers to the change of precipitation.

Third, the discrimination data were set to zero for both  $\delta D$  and  $\delta^{18}O$  since isotope fractionation would not occur during the process of plant water uptake. Run length performed in Markov Chain Monte Carlo (MCMC) was set to 'long'. Gelman-Rubin and Geweke diagnostic tests were employed to determine whether the model reached convergence before accepting the output of MixSIAR. The error structure and specify prior were set to 'Residual only' and 'Uninformative/Generalist' in the model, respectively. Previous studies have shown that oxygen isotope fractionation is small by comparison with hydrogen, so only  $\delta^{18}O$  was used in the data analysis using SPSS 22.0 and plotted using Origin 2018.

### 3. Results

#### 3.1. Precipitation

The precipitation in the study area was relatively adequate with obvious seasonal variations. The total precipitation in the study year (2020) was approximately 641 mm, 63 % of which occurred during the growing season, with the maximum precipitation in September (141 mm), accounting for 22 % of the total annual precipitation. Precipitation has a peak in August in the multiyear mean precipitation while peak in September in 2020. Plants growing season precipitation was significantly lower than the multi-year average. The average annual temperature in the last 15 years showed a significant increasing trend, with an average temperature increase of  $0.39^\circ\text{C}/10\text{a}$  (Fig. 2B).

During the study period, the isotopic composition of precipitation fluctuated widely (Table 2). There was a double-valley variation with time (Fig. 2c), and the two lowest values occurred in July and September, which was related to a large precipitation and low evapotranspiration. The slope and intercept of the local meteoric water line of growing season (LMWL-GS:  $\delta D = 7.28\delta^{18}O - 9.03$ ) were lower than those of the global meteoric water line (GMWL:  $\delta D = 8\delta^{18}O + 10$ , Craig, 1961), indicating that strong evapotranspiration occurred (Fig. 3).

#### 3.2. Isotopic composition of vegetation water

In the primary forest, the largest  $\delta^{18}O$  value was found in Ap. (Fig. 4), with a range of  $-9.42$  to  $-6.99$  ‰ and a mean value of  $-8.04 \pm 1.10$  ‰; the smallest  $\delta^{18}O$  value was found in Fm. with a range of  $-10.22$  to  $-9.17$  ‰ and a mean value of  $-9.56 \pm 0.47$  ‰. In the secondary forests, the largest  $\delta^{18}O$  values were in Bp. ranging from  $-9.09$  to  $-7.64$  ‰, with a mean value of  $-8.49 \pm 0.9$  ‰, and the smallest was in Qm. ranging from  $-10.17$  to  $-8.63$  ‰, with a mean value of  $-9.25 \pm 0.8$  ‰ (Fig. 4).

Most of the isotope values of xylem water in primary forests were similar to those of litter and soil water (Fig. 3a, b) and were distributed within the range of isotopic values of soil water from the litter layer to the 50 cm layer, indicating that soil water was the main source of plants in primary forests. However, in the secondary forest, the isotopic values of xylem water, especially Bp., were lower than the soil water, indicating that evaporation fractionation of soil water occurred (Fig. 3c, d).

#### 3.3. Isotopic composition of soil water

The  $\delta^{18}O$  of soil water in the primary forest and secondary forest decreased exponentially with increasing soil depth (Figs. 5, 6). The soil moisture content and  $\delta^{18}O$  value of shallow soil water in both forests fluctuated markedly, while the moisture content and  $\delta^{18}O$  in the middle and deep soil layers were relatively stable, which indicated that the surface soil water was greatly affected by rainfall, precipitation infiltration quantity and evapotranspiration. The soil water isotope values of both forest types were almost below the GS LMWL (Fig. 3), indicating that soil water came from precipitation and that evaporation occurred in the process of rainfall infiltration into the soil water. The slope and intercept of the soil water line in secondary forest (SF-SWL:  $\delta D = 3.188\delta^{18}O - 48.82$ ,  $R^2 = 0.85$ ) were lower than those of the primary forest (PF-SWL:  $\delta D = 3.778\delta^{18}O - 47.78$ ,  $R^2 = 0.86$ ), with higher  $\delta D$  and  $\delta^{18}O$  (Table 2), demonstrating that the

Table 2

Isotopic variations of  $\delta D$  and  $\delta^{18}O$  of litter layer, soil and xylem water in primary and secondary forest.

|                   |                | Primary forest    |                  | Secondary forest   |                   |
|-------------------|----------------|-------------------|------------------|--------------------|-------------------|
|                   |                | Min–Max           |                  | Mean $\pm$ SD      |                   |
| Precipitation (‰) | $\delta^{18}O$ | –14.32 to –4.31   |                  | –8.23 $\pm$ 2.37   |                   |
|                   | $\delta D$     | –117.04 to –34.50 |                  | –65.61 $\pm$ 19.61 |                   |
| Litter layer (‰)  | $\delta^{18}O$ | –8.50 to –5.66    | –8.47 to –4.45   | –6.95 $\pm$ 1.28   | –6.77 $\pm$ 1.38  |
|                   | $\delta D$     | –80.90 to –65.98  | –80.78 to –69.57 | –74.00 $\pm$ 6.16  | –73.00 $\pm$ 5.67 |
| Soil (‰)          | $\delta^{18}O$ | –11.89 to –7.73   | –13.21 to –5.44  | –9.81 $\pm$ 1.20   | –9.16 $\pm$ 1.64  |
|                   | $\delta D$     | –95.65 to –69.64  | –90.21 to –60.72 | –84.02 $\pm$ 6.62  | –78.02 $\pm$ 7.05 |
| Plant (‰)         | $\delta^{18}O$ | –10.22 to –6.99   | –10.17 to –7.37  | –8.88 $\pm$ 0.80   | –8.87 $\pm$ 0.70  |
|                   | $\delta D$     | –87.48 to –71.12  | –90.94 to –68.52 | –81.67 $\pm$ 5.16  | –82.46 $\pm$ 4.94 |

evaporation of soil water in secondary forests was stronger than that in primary forests. Overall, the soil water isotope values were affected by the soil depth and plant composition (Figs. 5, 6). With increasing soil depth, the isotope values of soil water in both forest types showed a decreasing trend. In primary forest, the differences in  $\delta^{18}\text{O}$  values in shallow soil water (litter–10 cm) were highly significant with middle (10–30 cm) and deep soil water (30–50 cm) in the primary forest ( $p = 0.00 < 0.01$ ), while there was no obvious difference in  $\delta^{18}\text{O}$  between middle and deep soil water. In the secondary forest, the  $\delta^{18}\text{O}$  variation in the soil water gradually increased with the depth of the soil. The  $p$  values of water  $\delta^{18}\text{O}$  between the litter layer and shallow soil layer (0 cm–5 cm and 5 cm–10 cm) were 0.036 and 0.029 ( $< 0.05$ ), respectively. There was a significant difference between shallow and middle soil water ( $p = 0.013 < 0.05$ ) and a highly significant difference between shallow and deep soil water ( $p = 0.00 < 0.01$ ) in the  $\delta^{18}\text{O}$  value. The shallow layer soil water had a higher stable isotope value and larger fluctuation, which indicated that the influence of precipitation, evaporation and vegetation on the water  $\delta^{18}\text{O}$  values in the shallow soil layer was stronger than that in the deep soil layers.

### 3.4. Contribution of precipitation to soil water

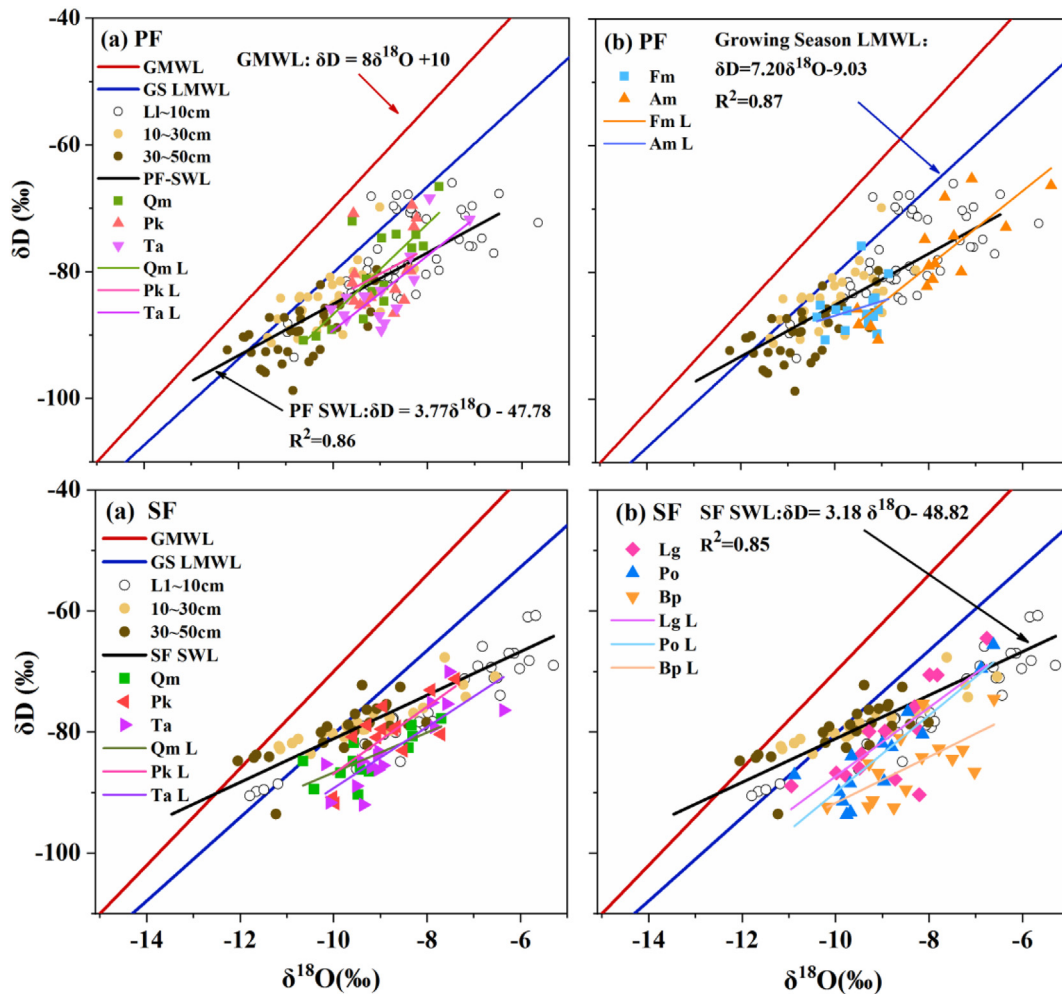
Spatially (Fig. 7), the soil water content of both forest types showed similar trends, first decreasing and then increasing with the soil depth. After

precipitation events, the soil water content of all layers in the secondary forest was lower than that in the primary forest. Temporally, there was a lag in the infiltration of precipitation into the soil. The maximum soil water content in each soil layer usually occurred on the second, third or fourth day after precipitation. The soil water content in the shallow soil layer (0 cm–10 cm) was more variable during the sampling period, while the deep layer (30–50 cm) was relatively stable.

Based on the MixSIAR model, the mean contribution of precipitation to soil water in the two forest types was significantly diverse (Fig. 8). When the precipitation was approximately 2.7 mm and 12 mm, the average contribution of precipitation to the soil moisture of the secondary forest was 26.19 % and 67.07 %, respectively, which was significantly higher than that of the primary forest (18.13 % and 42.96 %). However, after continuous precipitation events, the average contribution of precipitation to the soil moisture of the secondary forest (31.27 %) was lower than that of the primary forest (34 %).

### 3.5. Plant water use

Based on the direct comparison method and MixSIAR model, in primary forest, plants mainly used shallow soil water (L1–10 cm) when there was no or light rainfall (Fig. 5). The contribution ratio was 59.0 % (Qm.), 58.0 % (Pk.), 56.7 % (Ta.), 57.2 % (Ap.) and 45.0 % (Fm.), respectively (Fig. 9).



**Fig. 3.** The relationships between  $\delta\text{D}$  and  $\delta^{18}\text{O}$  in soil water (circles) and xylem water (including squares, triangles) during the sampling period. (a) and (b) denoted five xylems in primary forest include *Quercus mongolica* Fisch. ex Ledeb. (Qm), *Pinus koraiensis* Sieb. et Zucc. (Pk), *Tilia amurensis* Rupr. (Ta), *Fraxinus mandshurica* (Fm) and *Acer pictum* Thunb. ex Murray (Am). (c) and (d) denoted six xylems in secondary forest include Qm, Pk, Ta, *Larix gmelinii* (Rupr.) Kuzen. (Lg), *Populus davidiana* (Po) and *Betula platyphylla* Suk (Bp). GS GMWL referred to the global meteoric water line. GS LMWL referred to the local meteoric water line in growing season. PF SWL represented the soil water line based on isotope values of soil water in primary forest. SF SWL represented the soil water line based on isotope values of soil water in secondary forest. Qm L represented the Qm's xylem water line based on isotope values of Qm's xylem water.

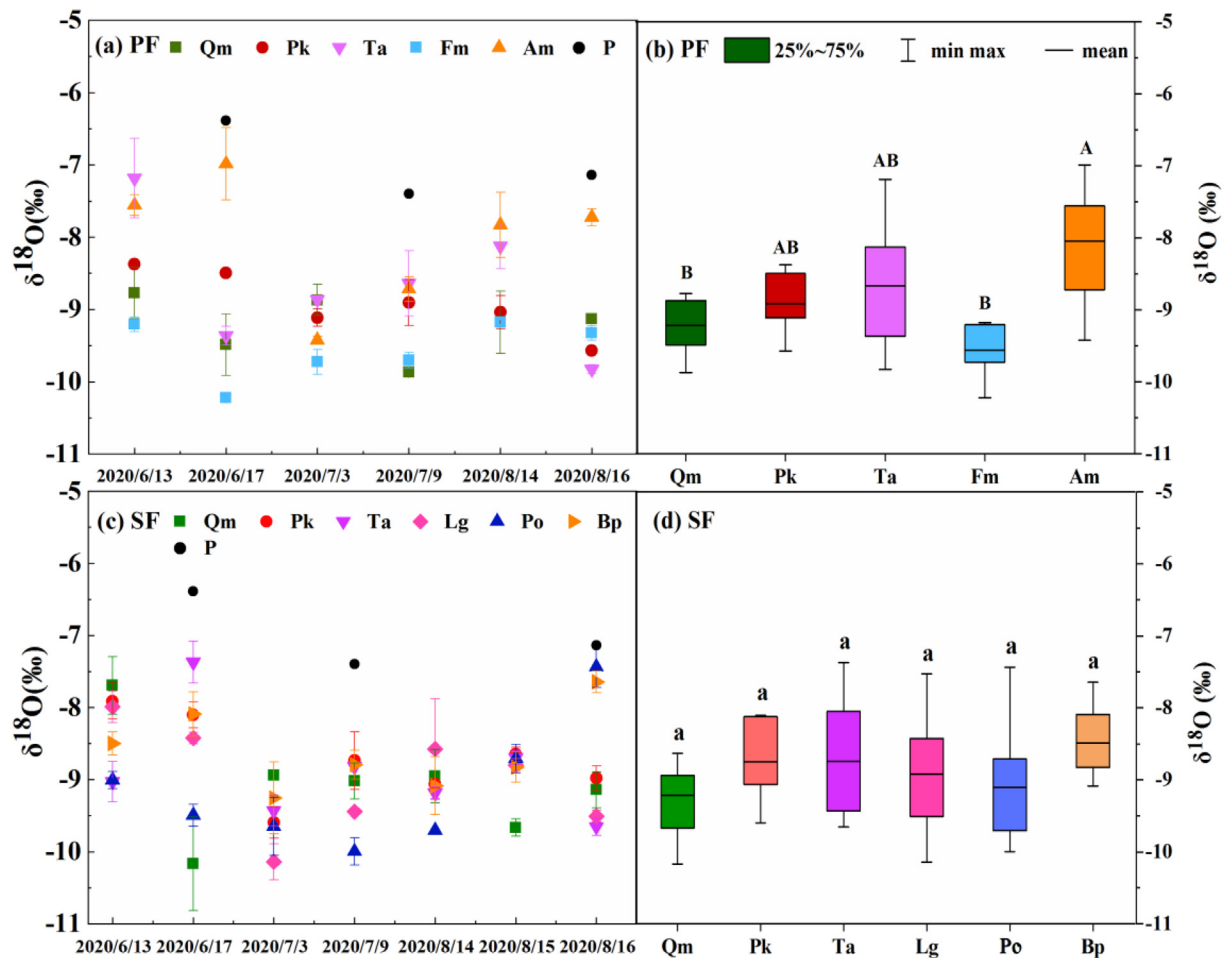


Fig. 4. Temporal variations in  $\delta^{18}\text{O}$  value of rainwater and xylem water in primary forest (PF) and secondary forest (SF). Significant differences were denoted by different lowercase letters ( $p < 0.05$ ) and extremely significant difference were denoted by different capital letters ( $p < 0.01$ ). P is the precipitation.

When there was middle rainfall, plants mainly utilized the soil water in the shallow and middle layers (10 cm–30 cm). After continuous precipitation (16.9 mm), all tree species except Ap. had multiple intersections with the middle and deep soil layers (30 cm–50 cm) (Fig. 5). These results showed that Pk. (Fig. 9b, g, l) and Ap. (Fig. 9e, j, o) mainly used the soil water in the shallow layer and Qm. switched water use between shallow and deep soil layers (Fig. 9a, f, k). Fm. and Ta. were changed greatly in their water use patterns. The intensity of precipitation dramatically influenced the plant water use pattern in the primary forest.

In the secondary forests, the primary water use was changed in the different soil layers and was not affected by the intensity of precipitation. When there was no precipitation, plants mainly used soil water in shallow layers (L1–10 cm) (Fig. 6). The water contribution of shallow soil layers to tree species was as follows: 47.5 % (Lg.), 45.3 % (Ap.), 45.0 % (Pk.), 42.5 % (Qm.), 41.7 % (Bp.), and 40.3 % (Po.). When there was light rainfall, the soil water in the middle and deep layers were mainly plant water sources (10 cm–50 cm). After continuous precipitation (16.9 mm), plants mainly used soil water in the middle layer (10–30 cm). The trend and change in plant water use in different tree species were very similar after precipitation events (Fig. 10), confirming that the species of secondary forest with similar plant water use patterns and independence on precipitation intensity.

## 4. Discussion

### 4.1. Water conservation capacity

Soil characteristics affect the soil water conservation capacity and then impact plant water use (Zhang et al., 2020). A higher soil water

conservation capacity usually results in a lower soil bulk density and higher total porosity (Zhang et al., 2021). In our study, the soil of the primary forest consisted of 12.3 % clay, 79.5 % chalk and 8.2 % sand, and the soil of the secondary forest consisted of 9.7 % clay, 69.5 % chalk and 20.8 % sand. This was consistent with the results of many studies that the vegetation gradually fine-grained the soil texture during the restoration process, and with the positive vegetation succession, the soil powder and clay content increased and the sand content decreased, where the clay content was highly correlated with the soil water conservation capacity (Peng, 2010; Sun et al., 2016). The soil of the primary forest and secondary forest could be divided into three categories according to the type of soil layer (Eggemeier et al., 2009; Wang et al., 2017): (1) shallow soil (litter to 10 cm): soil water isotopes and soil water content were susceptible to rainfall pulsation and evaporation with more drastic variation. (2) Middle soil (10–30 cm): a soil water regulating layer. Soil water isotopes and soil water content changed more moderately after precipitation for root distribution and humus decomposition (Cao et al., 2008). (3) Deep soil (30–50 cm): soil water isotopes and soil water content changed relatively stable. The soil bulk density increased with the soil depth (Table 3). However, the surface soil bulk density of the secondary forest was slightly smaller than that of the primary forest, while the total porosity and field water capacity of the deep soil in the primary forest were significantly larger than those in the secondary forest, indicating that the primary forest has a stronger soil interception capacity for precipitation than the secondary forest (Liu et al., 2021a).

The movement of soil water consisted of two opposite directions: upward convective transmission (evaporation) in the form of steam and downward movement in the form of molecular diffusion in the liquid phase (Zimmermann, 1967; Koeniger et al., 2016). When the water

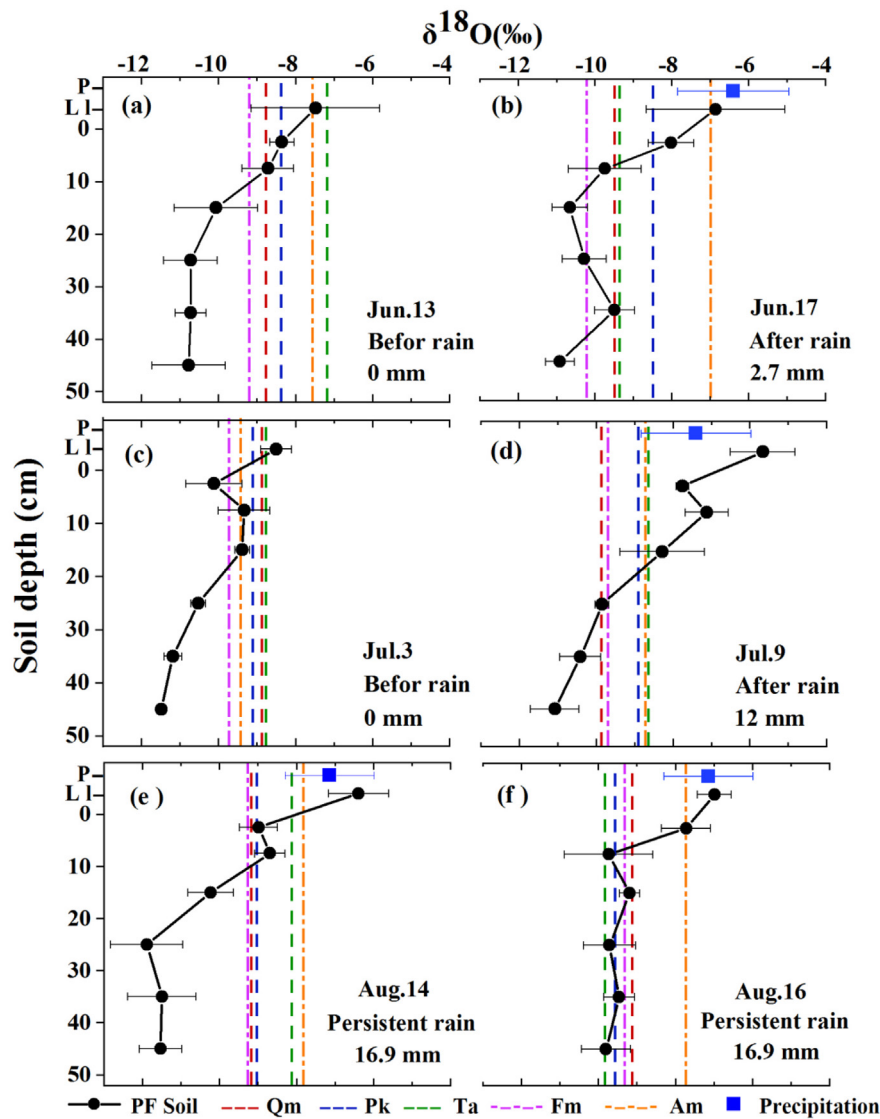


Fig. 5. Comparison of the soil water  $\delta^{18}\text{O}$  (L1–50 cm) and xylem water in primary forest after different precipitation events. Error bars represent standard deviation ( $n = 3$ ). (a) is Jun. 13 before rain, (b) is the Jun. 17 after rain with 2.7 mm, (c) is Jul.3 before rain, (d) is Jul. 9 after rain with 12 mm, (e) is Aug. 14, the first day during the persistent rain events, (f) is Aug. 16, last day during the persistent rain events.

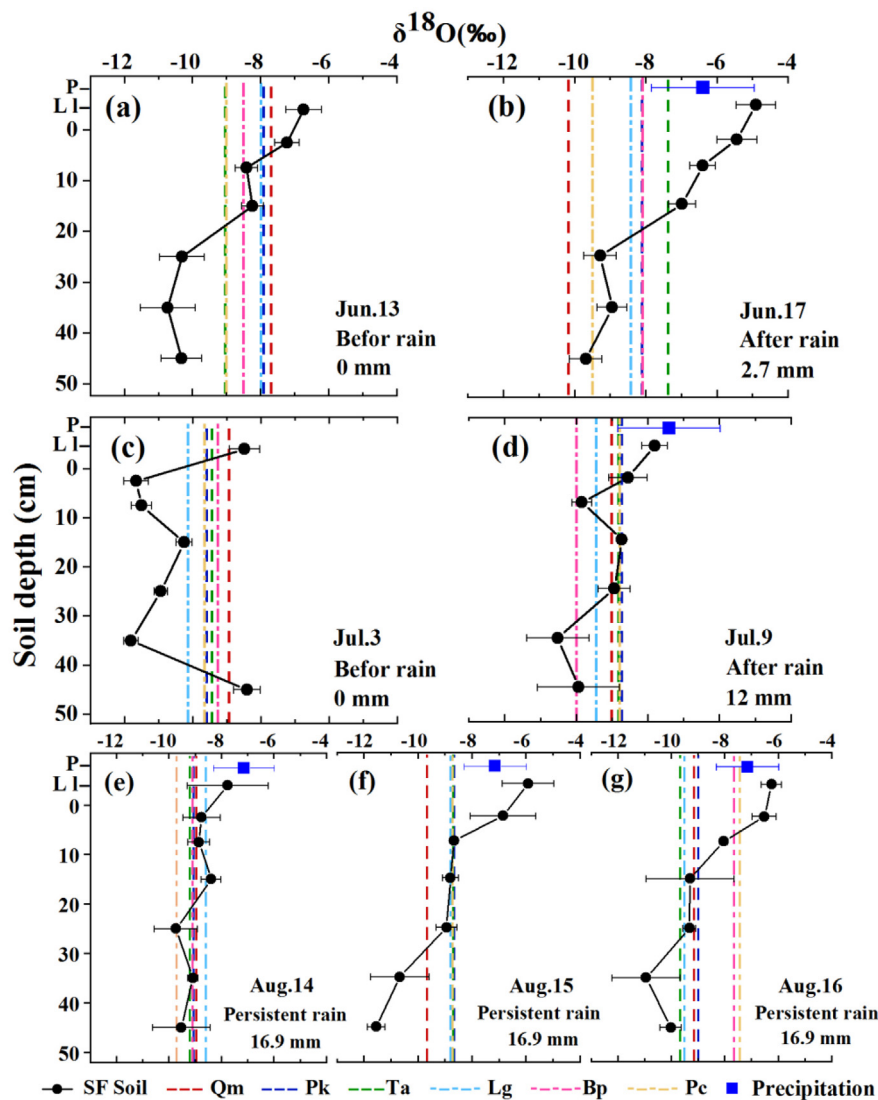
contents reached a balance in a certain soil layer, which was usually located within 20 cm of the soil surface,  $\delta$  was the highest and was called an “isotopic front” (Barnes and Allison, 1988; Hsieh et al., 1998). In the primary forest, when there was light rain, no isotopic front for the interception of canopy and litter. The Qm. mainly uses soil water in the shallow layer (Fig. 9a), while Ta. and Fm. still absorb water evenly in many soil layers (Fig. 9c, d). When there was mid-rain, the isotopic front appeared at the depth of the 5 cm–10 cm soil layer with a higher  $\delta^{18}\text{O}$  value in all soil layers. These results indicated that precipitation compressed the old soil water by piston flow in the 10 cm soil layer and rapidly infiltrated the 50 cm soil layer by preferential flow (Gazis and Feng, 2004; Zhao and Wang, 2020). At this time, the precipitation in the soil was dominated by infiltration and was less mixed with the deep soil water, Qm. used the deep soil water infiltrated by preferential flow (Fig. 9f), while Ta. and Fm. mainly used shallow soil water infiltrated by piston flow (Fig. 9h, i). During continuous precipitation, the isotopic front disappeared in the 5 cm–10 cm soil layer (Fig. 5f). The contribution of rainfall to the soil was equivalent in the 10 cm–50 cm soil layers, with a stable  $\delta^{18}\text{O}$  value and higher soil moisture content (Fig. 8c), indicating that the rainwater and soil water had been fully mixed. In the initial stage of continuous rainfall, Qm. mainly used shallow soil water. With increasing soil moisture, it

utilized multilayer soil water during the late stage of continuous precipitation. This result showed that Qm. could adapt to different precipitation patterns with wide ecological plasticity.

In the secondary forest, the plant water use pattern was hardly affected by precipitation intensity and the soil water flow pattern, but the situation of soil water was dramatically different from that in the primary forest. Before precipitation, there was an isotopic front located within the 10 cm–20 cm soil layer for intense evaporation (Fig. 6a, c, e). However, when there was light rain, the  $\delta^{18}\text{O}$  from the litter layer to the 30 cm soil layer significantly changed due to the weak interception of the canopy, and the isotopic front disappeared (Fig. 6b). When there was mid-rain (Fig. 6d), the precipitation rapidly infiltrated the 50 cm soil layer by preferential flow with a significant change in the  $\delta^{18}\text{O}$  value. During continuous precipitation, the isotopic front in 10 cm–20 cm soil layer disappeared (Fig. 6e, f, g), but the  $\delta^{18}\text{O}$  value changed dramatically in 30 cm–50 cm soil layer, combined with soil moisture content (Fig. 7) and precipitation contribution to soil water (Fig. 8f), it could be concluded that precipitation cannot be effectively mixed with deep soil water, and adequately supply to soil bound water leading to poor water conservation capacity of second forest.

Therefore, when there was light rain ( $\leq 10$  mm), the soil moisture content of the primary forest was higher than that of the secondary forest. In





**Fig. 6.** Comparison the  $\delta^{18}\text{O}$  value of soil water (L1–50 cm) and xylem water in secondary forest after different precipitation events. SF Soil denotes the average value of soil water. Error bars represent standard deviation ( $n = 3$ ). (a) is Jun. 13 before rain, (b) is the Jun. 17 after rain with 2.7 mm, (c) is Jul. 3 before rain, (d) is Jul. 9 after rain with 12 mm, (e) is Aug. 14, the first day during the persistent rain events, (f) is the Aug. 15, second day during the persistent rain events, (g) is the Aug. 16, last day during the persistent rain events.

contrast, the contribution of precipitation to the soil in the primary forest was lower than that in the secondary forest due to the higher interception capacity of the canopy and litter layer. This result was consistent with previous results (Brasil et al., 2018). In this study, rainwater infiltrated into the soil combined with piston flow and preferential flow in the primary forest, and preferential flow was the only form of flow in the secondary forest. Although the rainwater contribution to the soil water of the secondary forest was high (Fig. 8b, e), the soil water content of the primary forest in each layer was still significantly higher than that of the secondary forest (Fig. 7d, h). This result showed that precipitation could quickly infiltrate into the deep soil of the secondary forest, but the retention time was relatively short. When continuous precipitation (more than three days and the amount of precipitation >10 mm) occurred, the piston flow was dominant in the soil water flow of the primary forest, the water in the deep layer was fully mixed with 50 cm infiltration depth, and the time of water conservation capacity was relatively longer. In the secondary forest, the soil water flow was mainly preferential flow with shorter time of water conservation. It could be concluded that the forest composition and soil heterogeneity were the main factors leading to the differences in soil water flow and the water conservation capacity (Liu et al., 2021a).

#### 4.2. Plant water use pattern

As the burial depth of the groundwater level in the Changbai Mountains was approximately 10 m–60 m (Yan et al., 2018) and the plant roots in the study area are mainly distributed in shallow soil (Guo et al., 2016), it was relatively impossible for plants to utilize groundwater directly (Xu, 2020). There was also no irrigation or water inflow to the study area. Soil water is the main water source of plants with a thick litter layer and highwater conservation capacity (Fang et al., 2011). In the primary forest, the xylem isotopic values were significantly different in Ap., Qm., and Fm. ( $p < 0.01$ ) (Fig. 4b), indicating that the three plant species mainly used different soil layer moisture and there was no interspecific water use competition. Qm. were occupied a dominant position in positive community succession (Tian et al., 2021) with a strong ecological plasticity, its water source between shallow and deep soil water (Zhang et al., 2019b). Fm., a key tree species, has formed different ecological niche of water use (Tang et al., 2019b). In addition, there was no obvious difference ( $p > 0.05$ ) in  $\delta^{18}\text{O}$  of Qm., Pk., and Ta., implied that the water use of the three tree species was less influenced by the forest succession process. In contrast, there was no significant difference ( $p > 0.05$ ) in the xylem water isotopic values



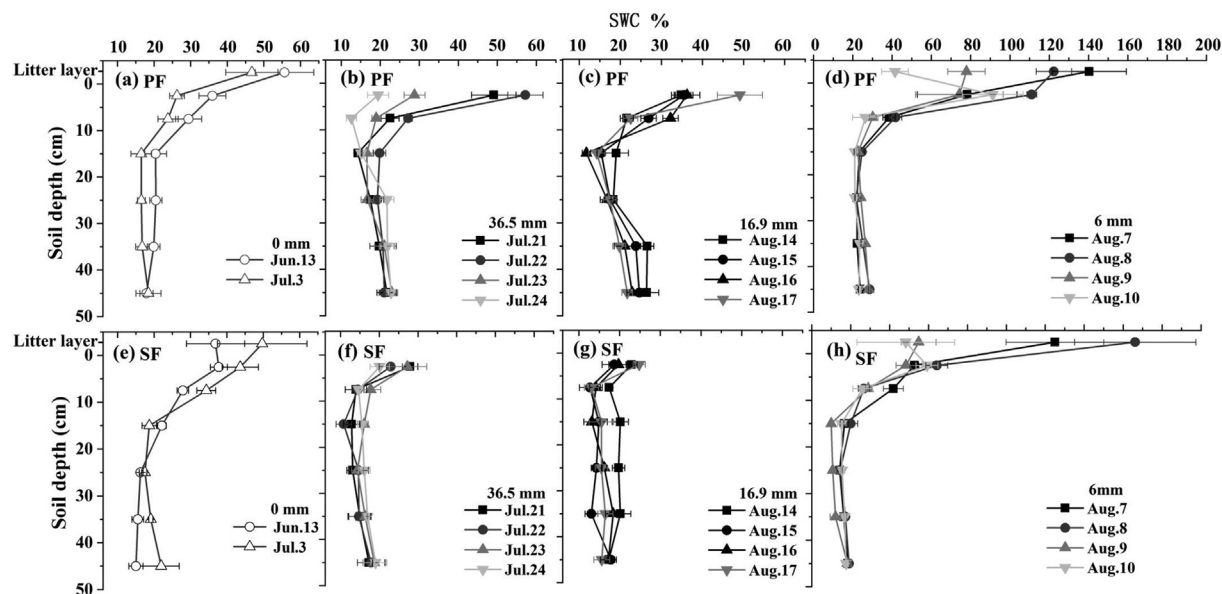


Fig. 7. Temporal variation of soil water content in primary forests (PF) and secondary forests (SF) after different amounts of precipitation (a–d for PF and e–h for SF). The SWC is the soil water content.

of the six tree species (Fig. 4d), showing that the soil water utilized by each tree species was from the same or similar soil layers, which indicated that the competition among secondary forest species was fierce in the water shortage period (Tang et al., 2019a; Zhao et al., 2021). The  $\delta^{18}\text{O}$  values of plant water in both stands showed a decreasing trend from June to July and an increasing trend in August (Fig. 4a, c), which was consistent with the trend of precipitation  $\delta^{18}\text{O}$  (Fig. 4), indicating that precipitation intensity influenced plant water use patterns to some extent and that plants responded to different levels of precipitation events.

#### 4.3. Soil reservoir hypothesis

There are several hypotheses on plant water use pattern. Brooks (2009) firstly proposed “two water worlds” that soil water in a watershed is compartmentalized into mobile and soil-bound water in the ecohydrological cycle, the soil-bound water is only available to plants and does not exchange with mobile water. However, recent work by Hervé-Fernández et al. (2016) in a rainy temperate zone in Chile, and by Mc Cutcheon et al. (2016) in a semi-arid, snow-dominated landscape in Idaho

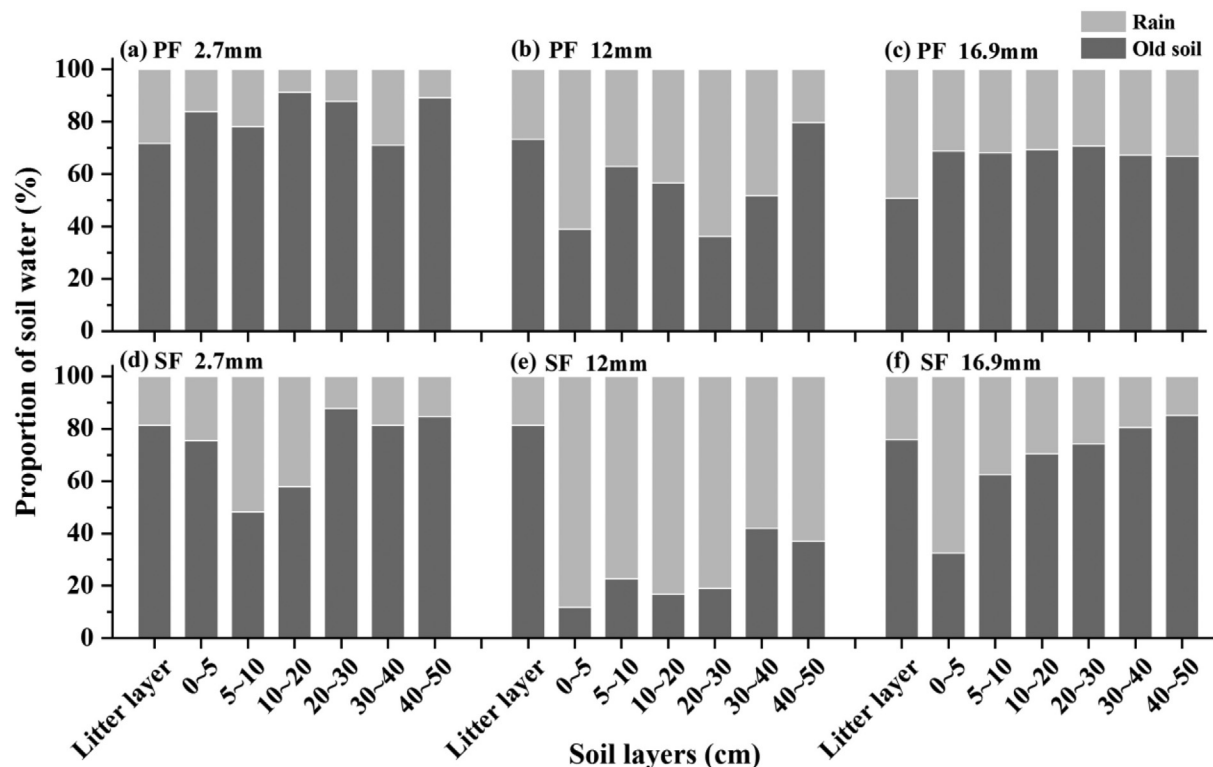
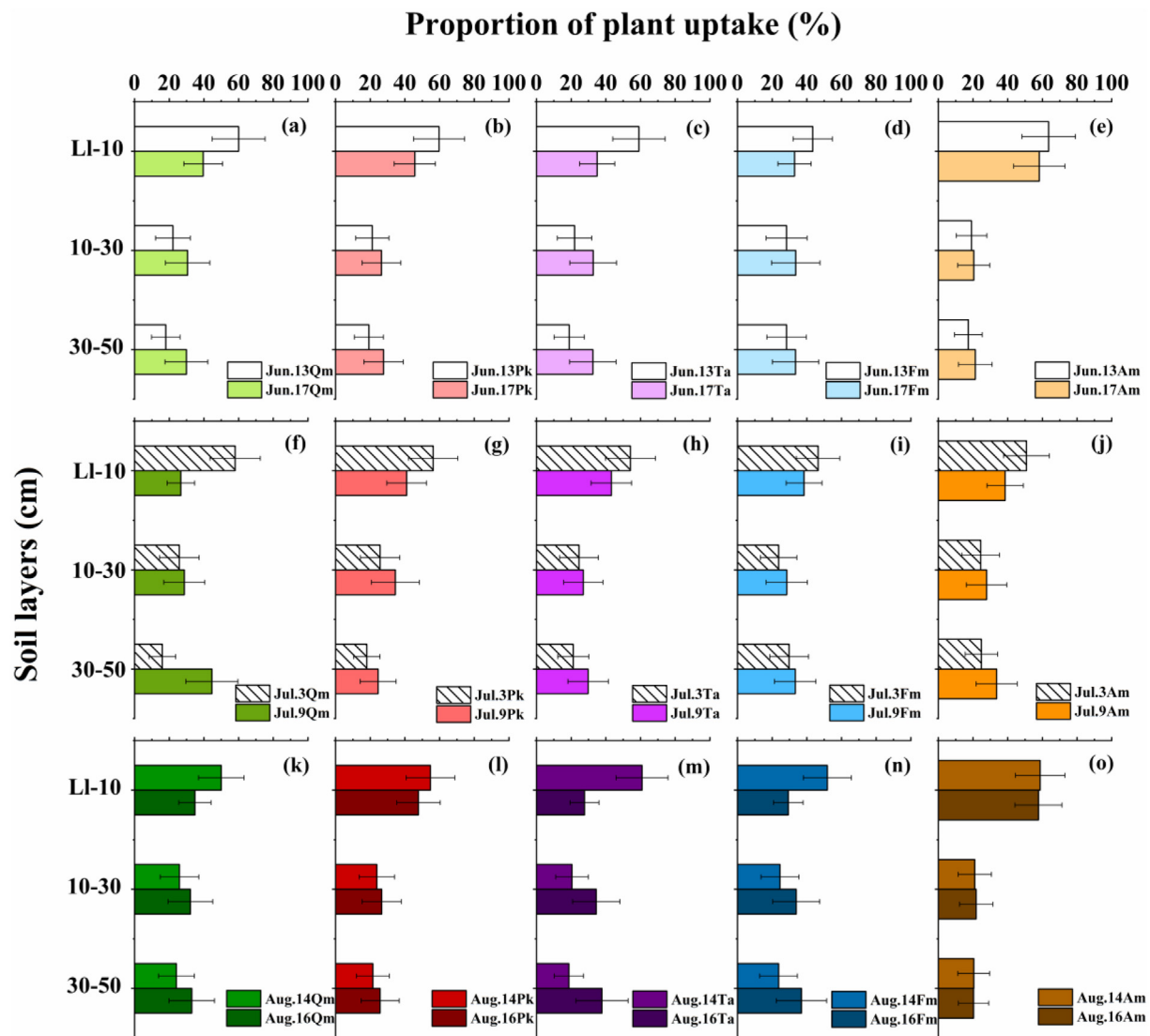


Fig. 8. Contribution of precipitation to soil water of primary forest (PF) and secondary forest (SF) after different precipitation.



**Fig. 9.** Plant water use variations in different soil layers based on MixSIAR in primary forest. The white square represents no precipitation events, and the color (from light to dark) denotes the change of precipitation amount. (a) to (o) are the different water use pattern, five images in each row are different water use patterns for the five species, the three images in each column are the water use pattern of a species in three different precipitation events.

demonstrated that during wet periods when soils are replenished there was sufficient mixing of water stored in the larger and smaller pores of the soil to infer that soil water was not clearly separated into two compartments.

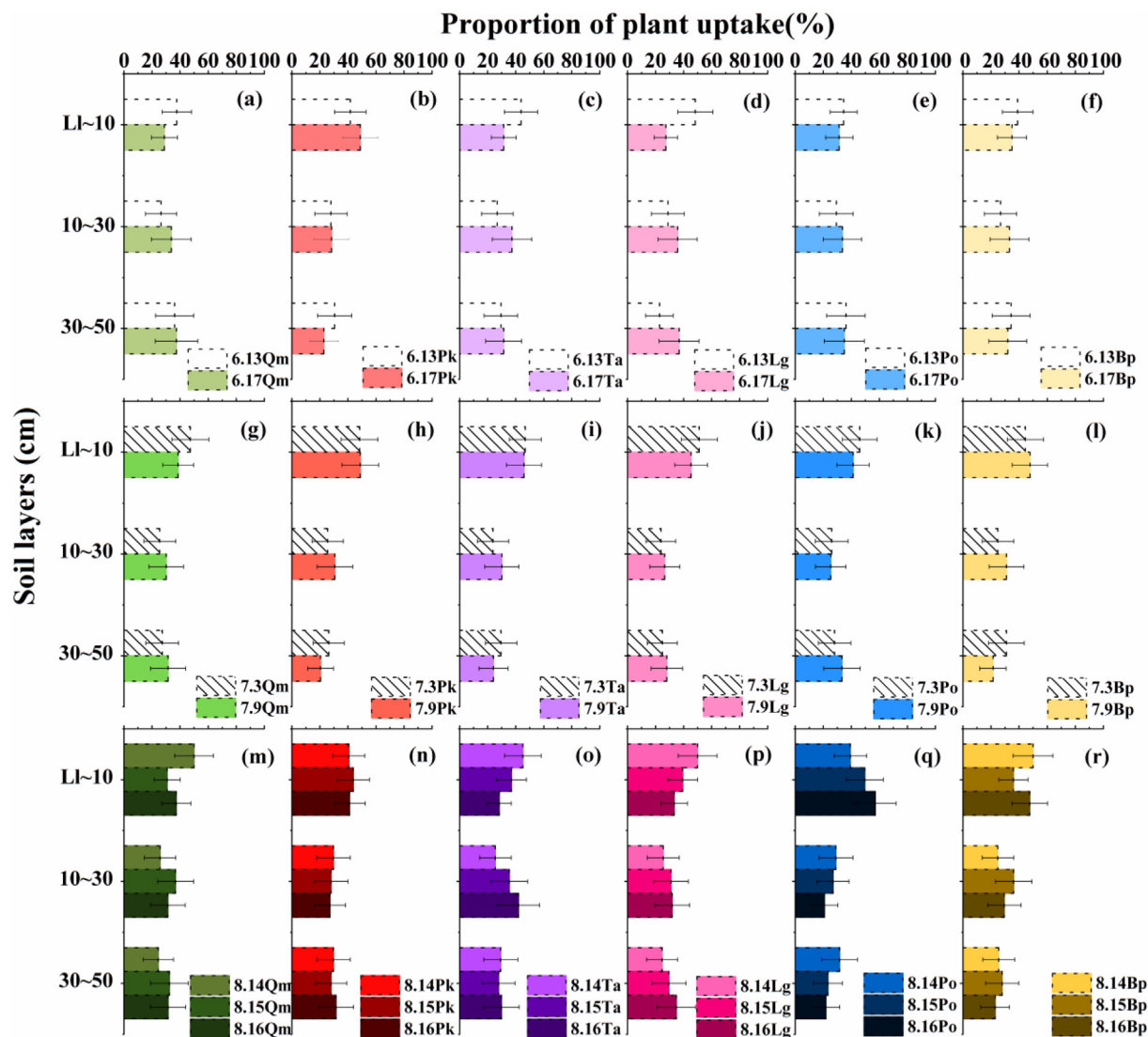
In secondary forests, precipitation enters the soil mainly in the form of preferential flow, which is the premise of ecohydrological division. Most soil water isotope values of secondary forest were located below the LMWL (Fig. 3c, d). The soil water line (SF-SWL) deviated from the LMWL, indicating that precipitation was not thoroughly mixed with soil water, and the soil water was still mainly “old water”. Moreover, the xylem water line deviated from the soil water line (SF-SWL) with a smaller slope, indicating that the plants preferentially and mainly utilized the soil “old water”, which was distributed in the small pores of the soil and enriched by evaporation. When the bound water of the soil decreased and was insufficient to fulfill the demand of plants, plants use the mixed water of rain and bound water, which confirmed the hypothesis of “two water worlds” (Bowling et al., 2017; Evaristo et al., 2015; Sprenger et al., 2019).

After the precipitation entered the soil in the primary forest, water mainly moved downward in the form of piston flow, leading the “new water” to thoroughly mix with the “old water” and slowly infiltrate deeper layers. At the same time, different tree species formed a stable ecological

niche of water use, and some tree species adjusted the water utilization strategy according to the precipitation intensity, which was more in line with the theory of a soil reservoir (Geris et al., 2015; Geris et al., 2017). Therefore, it could be concluded that soil texture and particle composition not only affect the movement mode of precipitation but also influence the segmentation of soil reservoirs (Brooks et al., 2010).

## 5. Conclusions

In this study, we analyzed the temporal and spatial changes in isotopic composition and water conservation capacity of two forest types in the Changbai Mountains and quantified the water use pattern of dominant species. The results could provide the following insights for forest restoration and management. First, the water conservation capacity of the primary forest ecosystem was higher than that of the secondary forest ecosystem. Under the background of global warming, compared with secondary forests, primary forests have a stronger regulatory ability for extreme droughts and floods, which could play an important role in ensuring the ecological security of the middle and lower reaches of rivers. Second, the dominant tree species in the primary forest formed a separate water use niche through healthy competition, which not only played a certain role in regulating



**Fig. 10.** Plant water use variations in different soil layers based on MixSIAR in secondary forest. The white square represents no precipitation events, and the color (from light to dark) denotes the change of precipitation amount. (a) to (r) are the different water use pattern, six images in each row are different water use patterns for the six species, the three images in each column are the water use pattern of a species in three different precipitation events.

flood disasters but also maintained the stability of forest ecosystems. It was suggested that proper management and multispecies allocation should be considered in the restoration of degraded forests or the construction of artificial forests. Third, the soil water in the primary forest was in line with the theory of a soil reservoir. In contrast, the soil water in the secondary forest was in accordance with the hypothesis of “two water worlds”.

Changbai Mountain has the most complete vertical zone of vegetation in East Asia. In this study, we only researched the zonal vegetation and did not deal with the vertical zone of vegetation. In the future study, the

researches about plant water use with the change of altitude and water circulation through precipitation, evapotranspiration and vegetation in Changbai mountain would be strengthened. And an ecological hydrological process model of Changbai Mountain area should be constructed to simulate and evaluate the variation of water conservation in climate sensitive area.

#### CRediT authorship contribution statement

Zeyu Sun and Mingming Feng: Methodology, Formal analysis and Writing; Xinyan Zhang and Wenguang Zhang: Conceptualization, Methodology and Writing-Review & Editing; Shaoqing Zhang, Yang Li, and Yiqiang Huang: Investigation; Wenjuan Wang: Supervision; Peng Qi: Resources; Yuanchun Zou and Ming Jiang: Project administration and Funding acquisition. All authors discussed the results and contributed to the preparation of the manuscript.

#### Data availability

The authors do not have permission to share data.

**Table 3**  
Soil properties in two forest types.

| Forest type           | Soil horizon (cm) | Sand content (%) | Bulk density (%) | Field capacity (g/kg) | Total porosity (%) |
|-----------------------|-------------------|------------------|------------------|-----------------------|--------------------|
| Primary forest (PF)   | 0–10              | 13.58            | 0.86             | 647.22                | 4.14               |
|                       | 10–30             | 6.32             | 1.35             | 322.43                | 4.27               |
|                       | 30–50             | 4.67             | 1.33             | 357.98                | 5.06               |
| Secondary forest (SF) | 0–10              | 23.80            | 0.71             | 646.36                | 5.51               |
|                       | 10–30             | 18.44            | 1.31             | 262.33                | 3.71               |
|                       | 30–50             | 20.20            | 1.52             | 191.79                | 1.50               |

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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